

Full Length Article

MEDiT: A mask-enhanced diffusion transformer model for fetal heart rate signal generation

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ARTICLE INFO

Keywords:

Data augmentation
Fetal heart rate (FHR)
Diffusion transformer
Mask-based diffusion model
Generative model

ABSTRACT

Fetal Heart Rate (FHR) signals are widely used in fetal monitoring and maternal-fetal health assessment. However, their acquisition is constrained by ethical limitations, motion artifacts, and signal loss, resulting in limited data availability for intelligent clinical analysis. To address these, this paper proposes a deep generative Data Augmentation (DA) framework for FHR synthesis. Specifically, we propose a Mask-Enhanced Diffusion Transformer model (MEDiT) to model various missing data scenarios and generate high-fidelity FHR signals in data-scarce clinical settings. First, a mask-enhanced reinforcement mechanism is designed to randomly mask signal segments at varying positions and lengths, forcing the model to reconstruct targeted regions from unmasked regions containing clinically critical information and thereby learn intrinsic physiological patterns under partial observability. Next, a Diffusion Transformer (DiT)-based generative architecture with a dual-branch learning strategy is developed to adaptively adjust the optimization objective under different masking ratios, improving both global temporal coherence and local waveform fidelity. Owing to self-attention-based global interaction and the threshold-guided branch design, MEDiT is able to capture long-range temporal dependencies while maintaining stable. Comprehensive experiments on a publicly available CTG dataset, including parameter sensitivity analysis, robustness evaluation, comparisons with state-of-the-art methods, downstream classification tasks, and clinician assessment, demonstrate the effectiveness of MEDiT. The results show that diffusion-based DA approaches perform better in both generation quality and training stability, and MEDiT further improves classification accuracy by 14.0% over the best baseline method (DDPM).

1. Introduction

Despite continuous improvements in maternal and neonatal care, approximately 2.3 million newborn deaths still occur worldwide annually. Both acute and chronic fetal distress remain the main factors in adverse outcomes, including neonatal asphyxia, disability, and death (Cnattingius et al., 2023). Timely identification of risks to the fetus and the pregnant individual has therefore become a central goal of prenatal monitoring (Marques et al., 2020). In clinical practice, cardiotocography (CTG) is widely applied in the late stages of pregnancy and during labor to assess fetal well-being (Amer-Wählin et al., 2001; Feng et al., 2021; Li et al., 2021). By recording Fetal Heart Rate (FHR) together with uterine contraction activity, CTG provides critical physiological and

pathophysiological information for obstetric decision-making. Among these, FHR is the most direct indicator of intrauterine fetal status and serves as a key basis for diagnosing fetal compromise (Melamed et al., 2021). With rapid advances in artificial intelligence, numerous FHR-based decision-support models, particularly data-driven deep learning approaches, have been developed and show promise for automated interpretation of CTG recordings.

However, enhancing the performance of these decision-support models generally requires large, clinically annotated datasets. In practice, data scarcity, missing values, and class imbalance hinder the construction of high-quality and balanced datasets. Additionally, the category distributions of the real clinical scenarios usually exhibit obvious long-tail distributions, i.e., head classes (normal fetuses) own sufficient

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<https://doi.org/10.1016/j.neunet.2026.108961>

Received 12 November 2025; Received in revised form 20 March 2026; Accepted 6 April 2026

Available online 13 April 2026

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samples whereas tail classes (pathological fetuses) are sparse and rare. Even widely used datasets contain only a limited number of pathological records. For instance, the CTG dataset from the Czech Technical University in Prague and the University Hospital in Brno (CTU-UHB) contains 552 records, of which only 40 are pathological cases (Goldberger et al., 2000; PhysioNet, 2014). It is clearly insufficient to meet the scale and diversity requirements of model training. Data collection is labor-intensive and costly, as it requires expert annotation, and involves strict ethical approval processes. Moreover, maintaining patient privacy is particularly important, as no organization or institution can directly access patients' personal information and clinical data without explicit consent. Therefore, an intuitive and effective solution is to generate enough high-quality and clinically meaningful synthetic FHR data to address the shortage of real, particularly pathological recordings.

In deep generative modeling, Generative Adversarial Networks (GAN) and Variational Autoencoders (VAE) are two main paradigms to address data scarcity and imbalance (Suh et al., 2021). They further include multiple variant architectures. On the one hand, Conditional GAN (CGAN) injects auxiliary conditions into both the generator and the discriminator, enabling class- or attribute-conditional synthesis (Mirza & Osindero, 2014). Deep Convolutional GAN (DCGAN) stabilizes training and enhances feature extraction through convolutional networks (Schirmer et al., 2017). However, their complex network structures are prone to unstable training and reduced sample quality. On the other hand, Conditional VAE (CVAE) augments VAE with conditional variables to steer the generation process, enabling data synthesis that satisfies specified conditions (Sohn et al., 2015). Temporal Difference VAE (TD-VAE) is a VAE tailored for sequence data, integrating the state-space model paradigm with temporal difference learning to handle complex, time-dependent sequences (Gregor et al., 2018). However, the resulting data are often weakly personalized and remain susceptible to diversity collapse (Yu et al., 2019). Furthermore, the normalizing flow achieves more stable training and faster convergence, yet the generated data are not superior to GAN or VAE (Kobyzev et al., 2020).

Diffusion Models (DM) and their variants have recently shown a strong ability to model complex data distributions and generate high-quality samples, particularly in image synthesis and video generation (Rombach et al., 2022; Yu et al., 2023). For example, Yu et al. developed a content-frame motion-latent diffusion model that introduces an autoencoder to encode videos into content frames and low-dimensional motion latents, thereby improving video quality and reducing computational cost (Yu et al., 2024a). Zhang et al. proposed a diffusion-4K method that applies wavelet-based fine-tuning on the Aesthetic-4K benchmark to generate highly detailed 4K images (Zhang et al., 2025). These advances provide new ideas for modeling complex biomedical signals such as FHR time series, an increasing number of studies are exploring diffusion-based approaches for time series modeling. For instance, TimeGrad combines autoregressive recurrent neural networks with diffusion for probabilistic forecasting of multivariate time series (Rasul et al., 2021), whereas TimeDiff employs a non-autoregressive conditional diffusion framework to reduce boundary discontinuities and improve long-term prediction (Shen & Kwok, 2023). Furthermore, the Diffusion Transformer (DiT) integrates the Transformer's global self-attention mechanism with the diffusion process, overcoming the inherent limitations of U-Net's local convolutions in modeling long-range dependencies. Such integration enables more effective conditional information fusion, improved scalability, and better pretraining transfer for complex signal generation. However, many DM-based methods presume fully observed signals during training, whereas sensor limitations and environmental factors routinely introduce missing segments in practice, creating a core challenge of modeling from incomplete signals while preserving performance.

In this paper, we focus on modeling a generative model for FHR signals in missing data scenarios. To address the above challenges, we propose a novel Mask-Enhanced Diffusion Transformer model (MEDiT), which combines a DiT-based generative backbone with a

Mask-Enhanced Reinforcement Mechanism (MERM) and a Dual-Branch Learning Strategy (DBLS) for adaptive learning under different missingness levels. The main contributions of this work are as follows:

- Most existing deep generative models rely on complete data distributions for training. To overcome this, this paper proposes a MERM to simulate incomplete signal acquisition in clinical practice. By guiding the model to reconstruct targeted regions from unmasked regions containing clinically critical information, MERM enhances the learning of intrinsic physiological patterns and improves generative modeling under varying missingness levels.
- To capture complex temporal dynamics in FHR time series, this paper proposes a DiT-based generative architecture with a dual-branch learning architecture. The DiT leverages self-attention to model long-range temporal dependencies, while the DBLS adaptively assigns distinct optimization objectives under low- and high-missingness conditions, guiding the model to learn complementary information from both masked and unmasked regions and thereby improving the overall generative modeling capability.
- Extensive experiments were conducted on the publicly available CTU-UHB dataset, including parameter sensitivity analysis, multi-scenario robustness analysis, comparisons with SOTA methods, and downstream classification evaluation. We further reproduced seven baseline DA methods for systematic comparison. The results demonstrate that MEDiT generates high-fidelity and physiologically plausible synthetic FHR data, and that improves downstream diagnostic performance.

The remainder of this paper is organized as follows. Section 2 reviews related work. Section 3 presents our proposed MEDiT method. Section 4 details the experimental setup and results. Finally, Section 5 concludes the paper with our findings and future work.

2. Related work

In this section, a review of current DA methods used for medical signals is presented, as well as basic information on DM and its variants.

2.1. Data augmentation for medical signals

Traditional Resampling Techniques. Oversampling expands minority class samples by duplicating existing data or performing simple operations such as random flips, scaling, or translations. In contrast, undersampling mitigates class imbalance by reducing majority class instances through clustering and feature extraction, such as random undersampling, cluster-based undersampling, and cost-sensitive learning (Taha et al., 2025). While these approaches are effective in dealing with class imbalance in general machine learning tasks, they are poorly suited to medical signal processing (Abdullah et al., 2025). Specifically, oversampling often produces highly similar samples that resemble direct data duplication, leading to redundancy and overfitting. Moreover, arbitrary geometric or temporal transformations can distort the physiological and pathological patterns inherent in medical signals, resulting in unrealistic synthetic samples (Alcaraz & Strodthoff, 2022; Giuffrè & Shung, 2023). Similarly, undersampling may discard substantial amounts of valuable information, causing underfitting and reduced generalization, which is particularly detrimental in data-scarce clinical settings (Yousefimehr et al., 2025). Consequently, recent studies have shifted toward deep learning-based generative models of learning latent signal distributions to generate high-qualification synthetic data.

Deep Generative Models. These approaches focus on learning complex latent distributions from real samples to generate high-fidelity synthetic data. For example, Zhang et al. integrated an auxiliary classifier to guide the generator in producing FHR signals conditioned on clinical categories (Zhang et al., 2022b), while Yu et al. extended the CGAN with self-attention and residual modules to better capture complex FHR

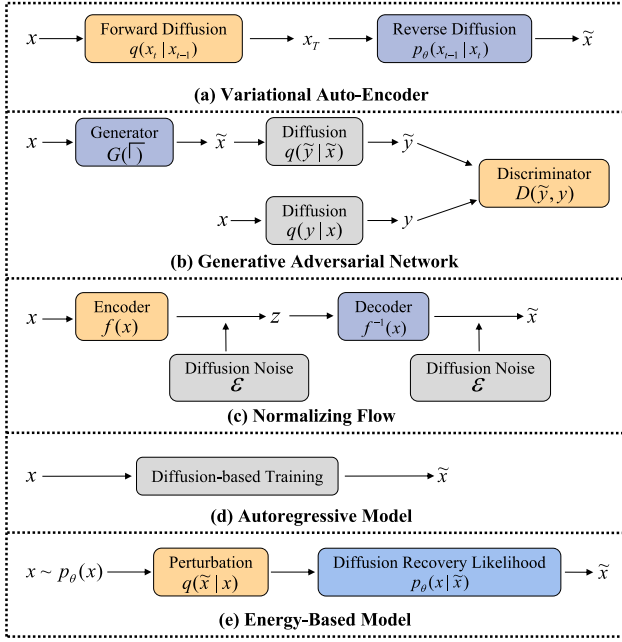


Fig. 1. Mainstream diffusion models and their training mechanisms in combination with other generative models.

dynamics information (Yu et al., 2024b). Similarly, Sang et al. and Luo et al. both developed specialized conditional VAE models to synthesize Electrocardiogram (ECG) (Sang et al., 2022) and Electroencephalography (EEG) (Luo et al., 2020) signals. And Bedin et al. modeled each ECG heartbeat as an independent morphological distribution sample and employed a diffusion model to learn the high-dimensional probability distribution of heartbeat morphology, achieving high-fidelity ECG reconstruction. Bedin et al. (2024). Wang et al. developed a DiffMDD which based generation model to characterize complex patterns in EEG data, an approach particularly effective for the high-dimensional and noisy characteristics of EEG signals (Wang et al., 2024). In which, GAN- and VAE-based methods remain prone to mode collapse, unstable training, and limited diversity of generated data. In contrast, DM reduces some of these issues through a stable and stepwise denoising process and has shown promise in generating high-quality data with greater diversity; its application to medical signal synthesis remains underexplored.

2.2. Diffusion-based generation approaches

As a typical probabilistic generative model, DM learns data distributions by progressively denoising Gaussian noise into structured samples, generates high-quality synthetic data. Since Ho et al. introduced Denoising Diffusion Probabilistic Models (DDPM) (Ho et al., 2020), DM and its variants have been widely applied in image synthesis (Lugmayr et al., 2022; Saharia et al., 2022), audio generation (Blattmann et al., 2023), and time-series forecasting (Huang et al., 2024; Shen et al., 2024; Tashiro et al., 2021). Due to their strong representation capacity and stable training ability, DMs have become one of the most versatile paradigms in deep generative modeling. Representative diffusion frameworks include DDPM, Denoising Diffusion Implicit Models (DDIM), conditional diffusion models, and hybrid architectures that combine diffusion with VAE, GAN, or Transformer (Awllla et al., 2025) (as illustrated in Fig. 1).

In time-series modeling, representative examples include DiffWave (Kong et al., 2021) for waveform generation, TimeGrad (Rasul et al., 2021) for probabilistic forecasting, and CSDI (Tashiro et al., 2021) for conditional time-series imputation. While these methods have shown promising performance for time-series generation, certain limitations re-

main. In particular, DiffWave may incur substantial sampling overhead, CSDI is primarily tailored to imputation and may depend on specific missing-pattern assumptions, and TimeGrad relies on an autoregressive backbone that limits parallel efficiency on long sequences. Hybrid diffusion frameworks have also been explored. For example, diffusion has been combined with bidirectional VAE to improve generative flexibility (Li et al., 2022), with GAN to stabilize adversarial training (Wang et al., 2023), and with autoregressive latent modeling to enhance diversity, which improves diversity but results in higher computational and memory demands (Gu et al., 2024).

These developments have further demonstrated the potential of diffusion-based generation for medical applications. DM-based approaches have been increasingly applied to medical signals (Li et al., 2024), including ECG (Bedin et al., 2024; Kwak & Jung, 2024) and EEG (Torma & Szegletes, 2025; Wang et al., 2024).

3. Method

In this section, we first present an overview of the proposed FHR generation model, MEDiT (Fig. 2), followed by a brief introduction to the DDPM framework and its training mechanism (Fig. 3). Then, describe the three core modules of MEDiT, including the MERM module, diffusion-based backbone network (Fig. 4), and the DBL strategy.

3.1. Overall architecture

The collected FHR signals often contain missing segments due to signal loss, maternal movement, or sensor displacement. To address this, we propose the MEDiT generative framework for synthesizing high-quality FHR signals under various missing-data scenarios, as illustrated in Fig. 2. Specifically, given an FHR sequence, a sliding window is first applied to divide it into overlapping segments $x \in \mathbb{R}^{N \times W}$, where N denotes the number of windows and W represents the window length. Next, a mask-enhanced reinforcement mechanism is introduced, which applies random masking in each training iteration to encourage the model to reconstruct masked regions from contextual information. The masked segments and their corresponding mask sequences are then fed into the DiT-based network to model long-range temporal dependencies, where a dual-branch learning strategy is designed to enhance generalization across varying levels of data missingness. Finally, the trained MEDiT generates synthetic FHR data conditioned on clinical labels and missing patterns, which are subsequently integrated with real data for downstream tasks.

3.2. denoising diffusion probabilistic model and its training mechanism

Unlike most deep generative models such as GAN, the core of DDPM lies in two Markov chains. The forward Markov chain gradually transforms the original data distribution into a pure Gaussian noise distribution, while the reverse Markov chain learns the mapping from the latent space back to the original signal space, thereby transforms the noise distribution into one that closely approximates real data. Specifically, the learning principle of DDPM is illustrated in Fig. 3.

Forward diffusion Process. Given data samples $x_0 \sim q(x)$ drawn from the underlying true distribution, the forward diffusion process progressively adds Gaussian noise to the data, generating x_t from its previous state x_{t-1} . Based on the chain rule and the Markov assumption, the conditional distribution $q(x_1, x_2, \dots, x_T | x_0)$ can be formulated as shown in Eq. (1).

$$q(x_1, \dots, x_T | x_0) = \prod_{t=1}^T q(x_t | x_{t-1}) \quad (1)$$

Where $q(x_t | x_{t-1}) = \mathcal{N}(x_t; \sqrt{1 - \beta_t} x_{t-1}, \beta_t I)$, it indicated that x_t is sampled from a Gaussian distribution with mean $\sqrt{1 - \beta_t} x_{t-1}$ and variance $\beta_t I$. Setting $\alpha_t = 1 - \beta_t$, $q(x_t | x_{t-1})$ can be derived as $q(x_t | x_{t-1}) = \mathcal{N}(x_t; \sqrt{\alpha_t} x_{t-1}, (1 - \alpha_t) I)$.

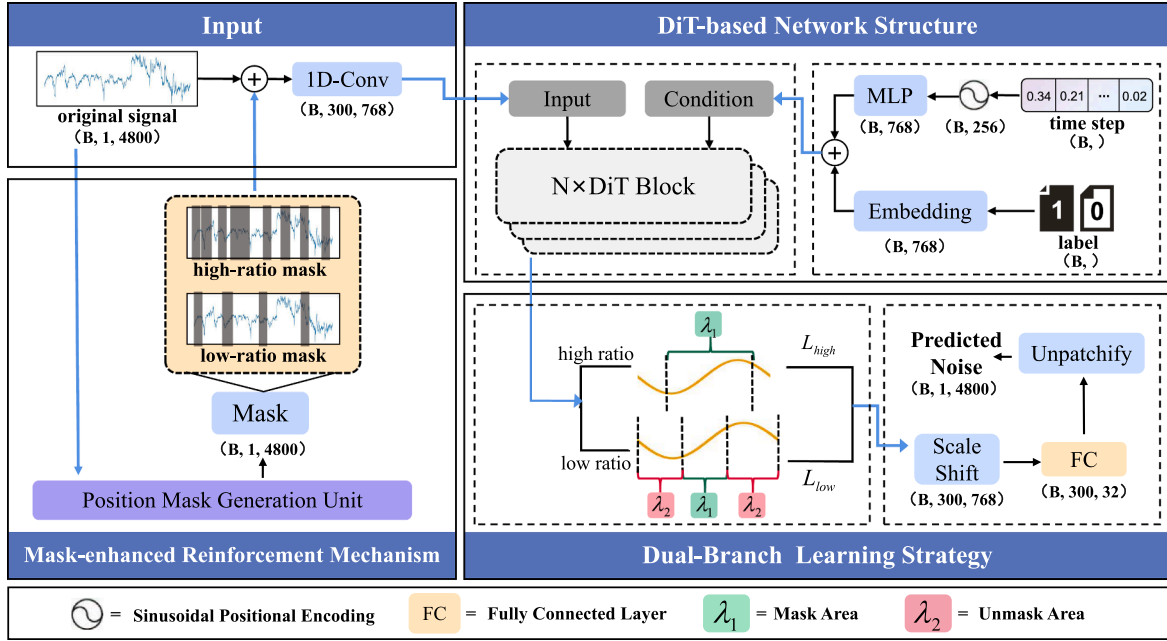


Fig. 2. Overview of the proposed MEDiT for FHR generation, including a mask-enhanced reinforcement mechanism, a DiT-based network structure, and a dual-branch learning strategy.

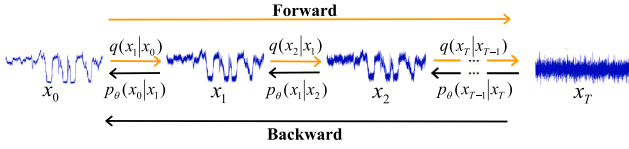


Fig. 3. The learning principle of DDPM.

Since x_t depends only on x_{t-1} , the relationship between x_t and x_0 can be derived based on the properties of the Markov chain. For $\varepsilon \sim \mathcal{N}(0, I)$, x_t can be expressed as:

$$x_t = \sqrt{\bar{\alpha}_t}x_0 + \sqrt{1 - \bar{\alpha}_t}\varepsilon \quad (2)$$

Where $\bar{\alpha}_t = \prod_{i=1}^t \alpha_i$.

Reverse denoising process. The reverse denoising process aims to reconstruct synthetic data that closely resemble the real data distribution from the input of Gaussian noise. Essentially, it involves training a noise prediction model $p_\theta(x_{t-1}|x_t)$ to learn the inverse process and thereby recover the original data. The reverse process can be parameterized as follows:

$$p_\theta(x_0, \dots, x_{t-1}|x_t) = p(x_t) \prod_{i=1}^t p_\theta(x_{t-1}|x_t) \quad (3)$$

$$p_\theta(x_{t-1}|x_t) = \mathcal{N}(x_{t-1}; \mu_\theta(x_t, t), \Sigma_\theta(x_t, t)) \quad (4)$$

Where $x_t \sim \mathcal{N}(0, I)$, $p_\theta(x_{t-1}|x_t)$ is assumed to be a normal distribution with learnable parameters, $\mu_\theta(x_t, t)$ and $\Sigma_\theta(x_t, t)$ are the mean and covariance matrix of the Gaussian distribution, respectively.

Objective function. The data sample x_0 arises through iterative sampling of the noise vector $x_{t-1} \sim p_\theta(x_{t-1}|x_t)$. Therefore, the training objective is to adjust θ , so that the joint distribution of the reverse denoising process $p_\theta(x_{0:T})$ approximates the forward diffusion process $q(x_{1:T}|x_0)$, measured via the Kullback-Leibler divergence:

$$L = \mathbb{E}q[D_{KL}(q(x_{t-1}|x_t, x_0)||p_\theta(x_{t-1}|x_t))] \quad (5)$$

which can be simplified to a weighted mean squared error loss between the true noise ε and the predicted noise $\varepsilon_\theta(x_t, t)$:

$$L = \min_\theta \mathbb{E}_{x_0, \varepsilon \sim \mathcal{N}(0, I)} \|\varepsilon - \varepsilon_\theta(\sqrt{\bar{\alpha}_t}x_0 + \sqrt{1 - \bar{\alpha}_t}\varepsilon, t)\|_2^2 \quad (6)$$

where q denotes the data distribution, $\varepsilon_\theta(x_t, t)$ is the parameterized neural network for noise prediction.

3.3. Design of mask-enhanced reinforcement mechanism

As the core component of MEDiT, the MERM is built upon a masking operation. Instead of explicitly filling missing values, it formulates dedicated masking schemes to emulate missing-data patterns commonly observed in clinically collected FHR sequences, thereby guiding the model to learn feature representations from targeted regions.

During clinical acquisition, unavoidable interferences such as sensor abnormalities, maternal motion, and fetal movement often induce local missingness in FHR signals. To replicate such real-world missing-data scenarios, MERM adopts a random positional masking strategy that selects a fixed proportion of time steps across the input sequence and marks them as missing, as shown in Eq. (7).

$$M_{ij} = \begin{cases} 1, & z_{ij} > r, \\ 0, & z_{ij} \leq r, \end{cases} \quad z_{ij} \sim \text{Uniform}(0, 1), \quad (7)$$

where r denotes the masking ratio, $M_{ij} = 1$ indicates that the corresponding entry remains observable, and $M_{ij} = 0$ indicates that it is masked.

The FHR generation process is then modeled as a sequence of local interpolation tasks. In each training iteration, MEDiT focuses on a designated region of the sequence and learns to recover informative features from that region before moving to another randomly selected region in the next iteration. Through this iterative process, MEDiT progressively learns the overall distribution characteristics of FHR signals, even in the presence of missing segments. As shown in Fig. 4(a), for the preprocessed input $X \in \mathbb{R}^{N \times W}$, each single-channel sequence is passed through MERM to construct a positional mask M under a prescribed masking ratio r , thereby generating a partially masked sample. Each masked sample is divided into a masked segment and an unmasked segment. The masked segment is then replaced with pure Gaussian noise, yielding the masked FHR sequence $\tilde{x}$. To simulate diverse missing-data conditions, we set multiple missing rates from 20% to 90% at intervals of 10%. The complete process is formulated in Eq. (8). Finally, The masked FHR

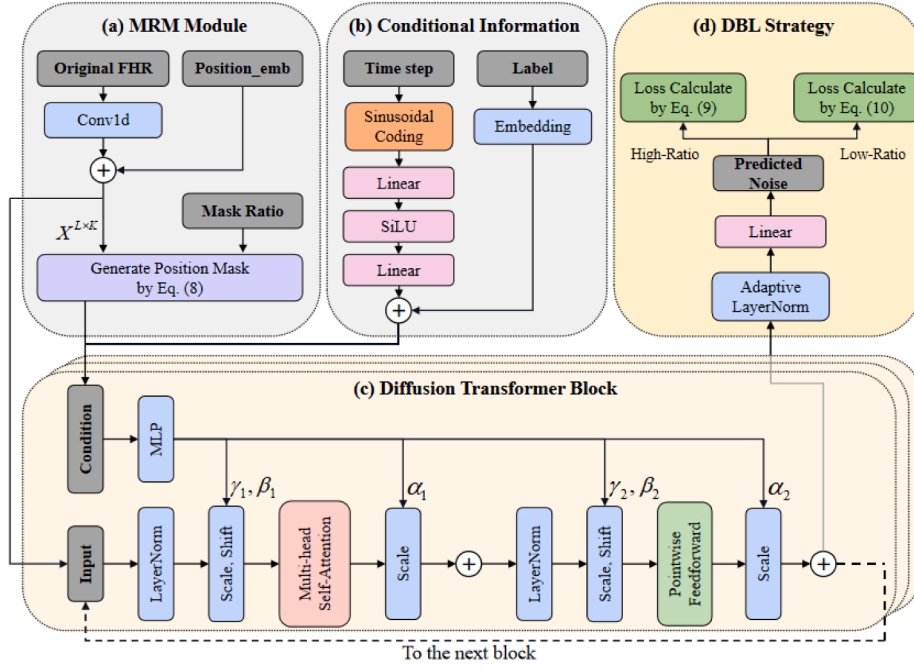


Fig. 4. Architecture of the proposed MEDiT. The backbone comprises 12 DiT blocks with 8 attention heads, a hidden dimension of 768, and a patch size of 16.

sequence $\tilde{x}$ serves as conditional input to the generation model.

$$\tilde{X} = M \odot X + (1 - M) \odot \epsilon, \quad \epsilon_{ij} \sim \mathcal{N}(0, \sigma^2), \quad (8)$$

where $\tilde{x} \in \mathbb{R}^{N \times W}$ denotes the masked FHR sequence, $M \in \{0, 1\}^{N \times W}$ is the positional mask generated by Eq. (7), $\odot$ represents element-wise multiplication, and $\epsilon \sim \mathcal{N}(0, \sigma^2)$ is Gaussian noise with standard deviation $\sigma = 1$.

This iterative masking strategy strengthens the model's ability to learn informative representations from partially observed FHR sequences and improves the quality of generated signals. By repeatedly reconstructing masked regions under varying missing ratios, MEDiT is encouraged to capture clinically relevant temporal structures, such as coherent baseline behavior and realistic variability patterns, which is aligned with the physiological interpretation principles emphasized in the FIGO guidelines (Melamed et al., 2021) and the international CTG consensus (Chandrasekaran et al., 2024).

3.4. diffusion transformer-based network structure

The noise prediction network plays a central role in diffusion-based generation. To model the complex temporal dynamics of FHR signals, we adopt a diffusion Transformer-based generative backbone, as illustrated in Fig. 4(b) and (c). Unlike traditional VAE- or GAN-based generative models, DiT integrates the DDPM with Transformer self-attention, enabling the model to jointly learn both global context and long-range temporal relationships within 1D time series. It is particularly important for FHR signals, whose physiological patterns often span multiple temporal scales rather than being determined only by adjacent time points. Specifically, the proposed network consists of three stages:

Stage I: Input Projection and Conditional Embedding. The Input Projection layer first patchifies the $\tilde{x}$ and appending the corresponding positional encoding, yielding latent representations that improve computational efficiency and representation compactness. Meanwhile, conditional information, including the diffusion timestep, clinical label and positional mask provided by MERM (Fig. 4 (a)) is encoded as guidance signals. Specifically, the diffusion timestep and clinical label are embedded through sinusoidal positional encoding and multilayer perceptrons, while the positional mask acts as an auxiliary conditioning tensor,

providing spatial cues about missing or masked regions in the input sequence.

Stage II: DiT Backbone Network. As its core are multiple stacked DiT blocks, each consisting of a multi-head self-attention layer and a point-wise feedforward network that together model long-range temporal dependencies. In this work, we design a modified DiT block incorporating adaptive layer normalization modulation (adaLN) and zero-initialized residual branches. It enables conditional embeddings to directly modulate normalization statistics, thereby improving expressiveness and accelerating training. The backbone consists of 12 DiT blocks with 8 attention heads, a hidden dimension of 768, and a patch size of 16.

Stage III: Output Projection and Reconstruction. After feature transformation by the DiT backbone, a linear projection layer decodes the latent representations back into the original feature space. The reconstructed tokens are then unpatchified to form a one-dimensional sequence, generating the noise prediction. This predicted noise is subsequently used in the reverse diffusion process to iteratively reconstruct high-quality FHR data with enhanced temporal smoothness. Subsequently, under the dual-branch learning strategy (Fig. 4 (d)), the discrepancy between the reconstructed and real FHR signals within the masked regions is analyzed and used as the optimization objective for further model refinement.

3.5. Dual-branch learning strategy guided generation

After feature transformation by the DiT backbone, a linear projection layer decodes the latent representations back into the original feature space, and the resulting patches are unpatchified to form a 1D noise prediction sequence. This predicted noise is then used in the reverse diffusion process to iteratively reconstruct high-quality FHR signals. The resulting predictions are further optimized under the dual-branch learning strategy (Fig. 4 (d)), which applies branch-specific supervision under different masking-ratio conditions, as detailed in Section 3.5

In MERM, we predefine positional mask ratios ranging from 20% to 90%. Based on these, DBLS divides the training process into two branches: a low-ratio branch (20%-60%) and a high-ratio branch (70%-90%). The detailed reconstruction loss are defined in Eqs. (9)-(10):

1) High-ratio branch: Since sufficient contextual information is available during training, the standard training procedure is retained, directing the model to focus on mask-specified regions. The optimization objective for this branch is defined as follows:

$$L_{high} = \|(X_T - X_0) \cdot M\|^2 + \lambda L_{VB} \quad (9)$$

Where L_{VB} represents the variational lower bound loss function, X_0 denotes the original ground-truth FHR signal, while X_T represents the FHR signal reconstructed through the reverse diffusion process. λ is the weighting coefficient, and M denotes the position mask, where 0 corresponds to masked areas and 1 to unmasked areas.

2) Low-ratio branch: Since large portions of the input remain unmasked, focusing solely on masked regions may lead to local optima. Therefore, we incorporate the unmasked regions into the reconstruction loss optimization, as defined in Eq. (10). This design enables the model to not only emphasize the mask-specified regions but also capture latent physiological features from the unmasked regions and integrate them into the generated data.

$$L_{low} = \lambda_1 \|(X_T - X_0) \cdot M\|^2 + \lambda_2 \|(X_T - X_0) \cdot EM\|^2 + \lambda L_{VB} \quad (10)$$

Where $EM = 1 - M$ indicates the positions of unmasked regions, and λ_1 and λ_2 are the weighting coefficients for masked and unmasked regions, respectively.

4. Experiments

This section presents the experimental evaluation of MEDiT for FHR generation. It first describes the dataset and experimental settings, followed by four experiments on publicly available benchmarks. The results show that MEDiT generates diverse, high-fidelity FHR signals and effectively enhances the performance of the downstream decision-support model.

4.1. Experimental settings

4.1.1. Dataset

This study used a publicly available CTG dataset from the CTU-UHB database, which contains 552 intrapartum recordings sampled at 4 Hz. The following preprocessing steps were as follows: **1) Data cleaning.** Records with extensive missing data or insufficient clinical information were excluded, accounting for approximately 10% of the dataset.

2) Label assignment. To address historical missing values in the remaining FHR recordings, we first applied the mini-batch-based sparse dictionary learning method described in our previous work (Zhang et al., 2021). The processed records were then categorized as normal ($pH \geq 7.15$ and $Apgar5/10 \in [9,10]$) or pathological ($pH < 7.05$), labeled as 0 and 1, respectively. All selected samples underwent secondary verification by a collaborating clinician to ensure the reliability of the classification boundaries.

3) Data processing. After preprocessing, the dataset contained 386 normal and 40 pathological samples. To obtain sufficient training data and construct a balanced dataset, we applied a sliding window with a 5 min stride to the pathological recordings, increasing their number from 40 to 280. We then randomly selected 280 normal samples to match the pathological class. Finally, all samples were standardized to a fixed length of 4800 time steps, corresponding to 20 min of FHR data at 4 Hz. Although MEDiT supports variable-length time-series inputs, fixed-length samples were used here to facilitate controlled training and evaluation. The final dataset therefore consisted of 280 normal and 280 pathological samples (560 in total). For the classification experiments, both normal and pathological samples were further divided into training, validation, and test sets at a ratio of 7:1:2, respectively.

4.1.2. Evaluation metrics

To comprehensively evaluate the generation performance of MEDiT, we conducted assessments from two aspects: quantify the discrepancy

between generated and real signals; and assess the impact of synthetic data on downstream classification tasks.

DM-specific evaluation. To evaluate distributional and pointwise discrepancies between generated and real data, four quantitative metrics are employed: Maximum Mean Discrepancy (MMD), Wasserstein Distance (WD), Fréchet Distance (FD), and Root Mean Squared Error (RMSE). MMD is a kernel-based nonparametric metric that measures the distance between two probability distributions in a Reproducing Kernel Hilbert Space (RKHS). WD, also known as Earth Mover's Distance, quantifies the minimum cost of transforming one probability distribution into another. FD measures the overall distance between two Gaussian approximations of the feature distributions and is effective for assessing distributional similarity in high-dimensional feature spaces. RMSE is a standard point-wise metric that quantifies the average magnitude of differences between generated and real data. These metrics are formally defined as follows:

$$\begin{cases} MMD(F, P, Q) = \sup_{\|f\|_H \leq 1} E_P[f(x)] - E_Q[f(y)] \\ WD(P, A) = \inf_{\gamma \sim \prod(P, Q)} E_{x, y \sim \gamma} [\|x - y\|] \\ FD(A, B) = \inf_{\alpha, \beta \in [0,1]} \max_{t \in [0,1]} \{d(A(\alpha(t)), B(\beta(t)))\} \\ RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (x - y)^2} \end{cases} \quad (11)$$

Where $\sup$ denotes the supremum; E_P represents the expectation under distribution P ; $f(\cdot)$ is the mapping function; and $\prod(P, Q)$ represents the set of all possible joint distributions obtained by coupling the two probability distributions.

Medical diagnosis-specific evaluation. The ultimate objective of data augmentation is to enhance the performance of downstream classification tasks. Accordingly, we employ an CNN-BiLSTM model (Liu et al., 2021) to evaluate the positive/negative impact of the augmented dataset on classification performance. Four evaluation metrics are used: Accuracy (Acc), Precision (P), Recall (R), and F1-score (F1).

$$\begin{cases} ACC = \frac{TP+TN}{TP+FP+FN+TN} \\ P = \frac{TP}{TP+FP} \\ R = \frac{TP}{TP+FN} \\ F1 = \frac{2TP}{2TP+FP+FN} \end{cases} \quad (12)$$

Here, TP, FP, TN, and FN represent true positives, false positives, true negatives, and false negatives, respectively.

4.1.3. Experimental platform

All experiments were conducted on an NVIDIA RTX 4060 GPU (8 GB memory) using Python 3.12 and PyTorch 2.2.2. The models were optimized with AdamW, initialized with a learning rate of 10^{-4} . Each training session consisted of 5000 iterations with a batch size of 16, taking approximately 14 h to complete. In low-missingness scenarios, λ_1 and λ_2 were set to 0.75 and 0.25, respectively.

4.2. Parameter sensitivity analysis

In this section, we conduct experiments to analyze the sensitivity of several hyperparameters and determine the optimal configuration for the proposed MEDiT. Four essential training parameters were examined: learning rate, batch size, number of epochs, and sample steps. Their effects were evaluated using MMD, WD, FD, and RMSE, as summarized in Fig. 5.

As shown in Fig. 5, the learning rate significantly impacts the quality of generated signals. Small learning rates (e.g., 10^{-5} and 10^{-4}) yielded consistently strong performance across all metrics, with low MMD, WD, FD, and RMSE values indicating high similarity to real signals. A similar trend was observed for batch size, where a value of 16 achieved the lowest scores across all four metrics and thus the best generation quality. Moreover, increasing the number of training epochs generally enhances generation performance. However, the gains became marginal

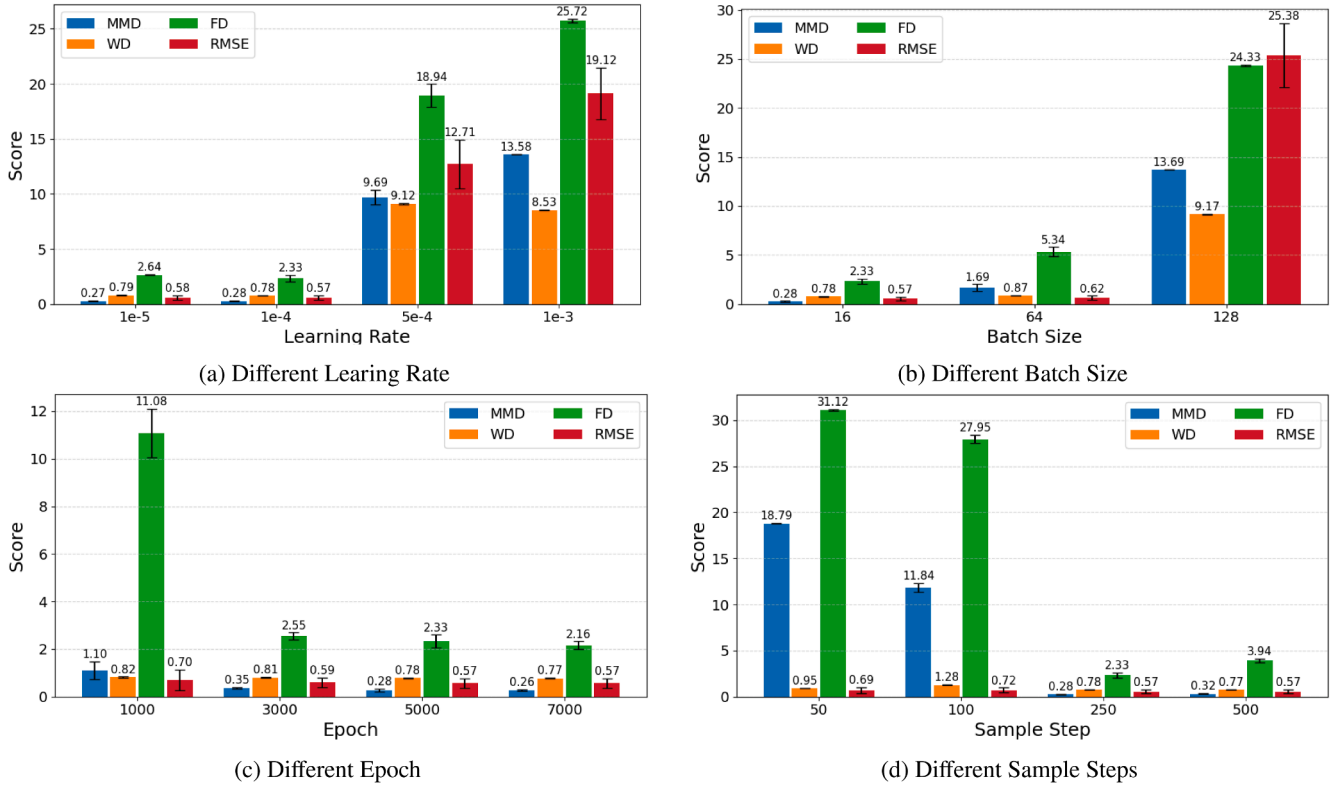


Fig. 5. Performance comparison under different parameter settings: (a) Learning rate; (b) Batch size; (c) Number of epochs; (d) Sample steps.

Table 1

Performance under different masking ratios and loss computation strategies.

Loss Strategy	Ratio	Metrics			
		MMD	WD	FD	RMSE
High-Ratio Loss on Low-Ratio Branch	20%	0.2829	0.7528	2.1753	0.4906
	30%	0.2716	0.7424	1.9985	0.5135
	40%	0.2875	0.7584	2.1839	0.5178
	50%	0.2486	0.7831	2.5196	0.5245
	60%	0.2828	0.7859	2.4041	0.5353
Low-Ratio Loss on High-Ratio Branch	70%	0.2616	0.7492	2.1378	0.5177
	80%	0.2537	0.7538	2.2890	0.5279
	90%	0.2570	0.7635	2.1452	0.5404

beyond 5000 epochs, indicating diminishing returns relative to the additional computational cost. Likewise, increasing the number of sampling steps improved temporal reconstruction quality and generation fidelity, but the improvements plateaued beyond 250 steps. Therefore, the final configuration was set to a learning rate of 10^{-4} , a batch size of 16, 5000 training epochs, and 250 sampling steps, balancing generation quality and computational efficiency.

In addition, within the DBLS framework, we divide the training process into a high-ratio branch and a low-ratio branch. Specifically, the masking ratio of the high-ratio branch ranges from 70% to 90%, while that of the low-ratio branch ranges from 20% to 60%. To validate the rationality of this partition, we designed a loss-swapping experiment in which the loss function of the high-ratio branch was applied to the low-ratio branch, and vice versa. This setting serves as a direct ablation of the branch-specific objective. As shown in Table 1, exchanging the loss computation strategies between the two branches consistently leads to a degradation in generation quality, which further confirms that the original ratio division and branch-specific optimization objectives are well aligned.

4.3. Multi-scenario robustness analysis

To evaluate the robustness and generalization ability of MEDiT, we conducted experiments under varying position mask ratios ranging from 20% to 90%, with an increment of 10%. The standard DiT structure served as a baseline, corresponding to the ablation setting where the MERM module was removed (i.e., position mask ratio = 0%).

Fig. 6 presents a comprehensive comparison of generation performance across different position mask ratios. Although minor fluctuations occur around mask ratios of 60% and 90%, the overall generation fidelity remains stable and consistently exceeds that of the baseline DiT. This can be attributed to the adaptive masking mechanism and the dual-branch learning strategy, which enable the model to effectively reconstruct temporal dependencies and preserve local structural details under different missingness levels. A representative examples of masked original signals and their corresponding generated signals under both low and high masking ratios is further shown in Fig. 7. Overall, the results demonstrate that MEDiT maintains consistent performance across multiple missing-data scenarios and outperforms the standard DiT, confirming its strong adaptability to incomplete inputs.

4.4. Comparative evaluation with SOTA methods

In this part, we reimplemented seven SOTA baselines and compared them with the proposed MEDiT. Specifically, the compared methods cover three categories: two traditional perturbation-based augmentation methods (Jittering and Scaling), three GAN-based generative models (CWGAN Zhang et al., 2022a, DCGAN Schirrmeister et al., 2017, and CTGAN Yu et al., 2024b), and two diffusion-based models (DDPM Ho et al., 2020 and CSDI Tashiro et al., 2021). The main results are summarized in Tables 2, 3, and Fig. 8.

Traditional augmentation methods introduce only limited perturbations to the original samples without learning their intrinsic data distribution. Although they achieve low MMD values, this mainly reflects

Table 2

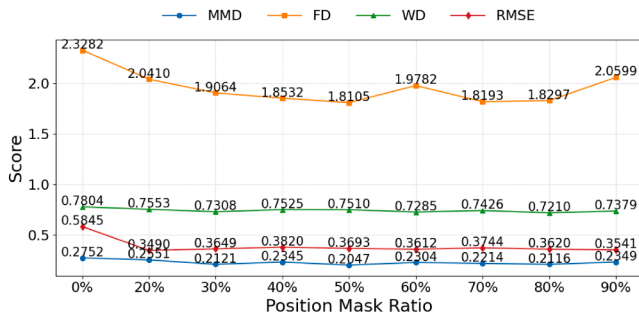
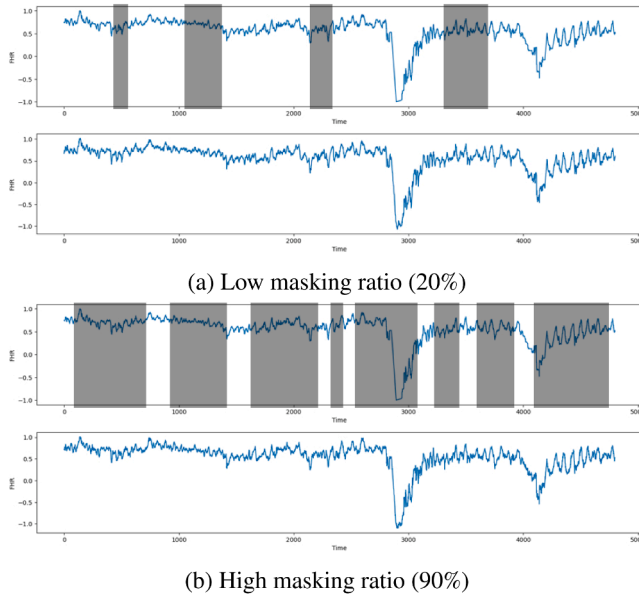
The key features and performance comparison of different methods. The best and second-best results are marked in bold and italic.

Model	Key Features				Test Performance			
	Deep Learning Model	Label Conditions	FHR-specific Generation	Support Missing Segments	MMD($\downarrow$)	WD($\downarrow$)	FD($\downarrow$)	RMSE($\downarrow$)
Jittering	$\times$	$\times$	$\times$	$\times$	0.3684	0.8598	4.6729	0.5947
Scaling	$\times$	$\times$	$\times$	$\times$	0.5114	0.8754	4.9698	0.5917
CWGAN	$\checkmark$	$\checkmark$	$\times$	$\times$	13.1707	0.8285	16.7373	0.4112
DCGAN	$\checkmark$	$\times$	$\times$	$\times$	4.3189	0.9781	31.1895	0.8556
CTGGAN	$\checkmark$	$\checkmark$	$\checkmark$	$\times$	4.0492	0.7776	10.4955	0.5883
DDPM	$\checkmark$	$\checkmark$	$\times$	$\times$	0.2888	0.7561	2.5036	0.5099
CSDI	$\checkmark$	$\checkmark$	$\times$	$\checkmark$	0.5860	0.7452	2.8581	0.5117
MEDiT-20%	$\checkmark$	$\checkmark$	$\checkmark$	$\checkmark$	0.2551	0.7553	2.0410	0.3490
MEDiT-90%	$\checkmark$	$\checkmark$	$\checkmark$	$\checkmark$	0.2349	0.7379	2.0599	0.3541

Table 3

Cost analysis of different methods.

	Jittering	Scaling	CWGAN	DCGAN	CTGGAN	DDPM	CSDI	MEDiT
FLOPs(G)	–	–	0.14	1.06	1.75	8.56	18.77	8.54
Parameters(M)	–	–	9.69	6.34	1.47	23.49	0.41	23.27
One-shot training time	–	–	0.3823	1.8341	0.3235	3.9274	3.6195	3.8325
Sampling time	9.4235	2.9891	0.0139	0.0139	0.0615	1.6790	2.0043	1.4342

**Fig. 6.** Performance comparison under varying mask ratios.**Fig. 7.** Generation examples under low (20%) and high (90%) masking ratios.

superficial similarity to the training data rather than genuine generative ability. In contrast, GAN-based approaches exhibit stronger generation ability, but their performance is often constrained by training instability and mode collapse, leading to inconsistent fidelity across metrics, particularly in MMD and FD. Diffusion-based baselines generally show better

robustness and generation quality than traditional and most GAN-based methods, and the proposed MEDiT outperforms all baselines across most of the metric, demonstrating its ability to synthesize high-quality FHR signals that closely resemble real physiological data.

In addition to generation quality, we further evaluated the computational efficiency of each method. Table 3 reports the FLOPs, parameter counts, training time, and sampling time of all compared approaches. As expected, GAN-based models require fewer parameters and are substantially faster in both training and inference than diffusion-based models. Although MEDiT incurs a moderately higher computational cost than GAN baselines, it delivers markedly superior generation quality. Moreover, its inference efficiency is slightly better than that of the standard DDPM framework.

Fig. 8 presents two generation examples from six deep generative models. MEDiT performs well on both normal and pathological FHR signals, suggesting strong generalization ability across diverse physiological conditions.

4.5. Classification performance evaluation

To further evaluate the effectiveness of MEDiT in addressing data imbalance for FHR classification tasks, we incorporated synthetic signals into the training set at varying augmentation ratios. From the CTU-UHB database, all of the 40 pathological and 160 normal FHR signals (imbalance ratio 1:4) were selected as the real data subset. The selected real signals were combined with synthetic pathological data generated by different DA methods, with imbalance ratio ranging from the original 1:4 to a balanced 1:1. Specifically, we incrementally added 0, 40, 80, and 120 synthetic pathological data to the original 40 real pathological samples, resulting in class ratios of 1:4, 1:2, 3:4, and 1:1, respectively. All data were partitioned into training, validation, and test sets at a ratio of 7:1:2, with the test set consisting exclusively of real data. To further ensure the reliability and privacy of the augmented dataset, all generated pathological samples were evaluated by collaborating clinicians to verify their clinical authenticity and compliance with established pathological criteria.

Fig. 9 presents a comprehensive comparison of classification performance across different DA methods and imbalanced ratios. To demonstrate MEDiT's robustness under missing-data scenarios, we examined two extreme masking configurations: 20% and 90% mask ratios, denoted as MEDiT-20 and MEDiT-90, respectively. The results demonstrate that all DA methods improve classification performance as the training data increases, with the degree of improvement varying across methods.

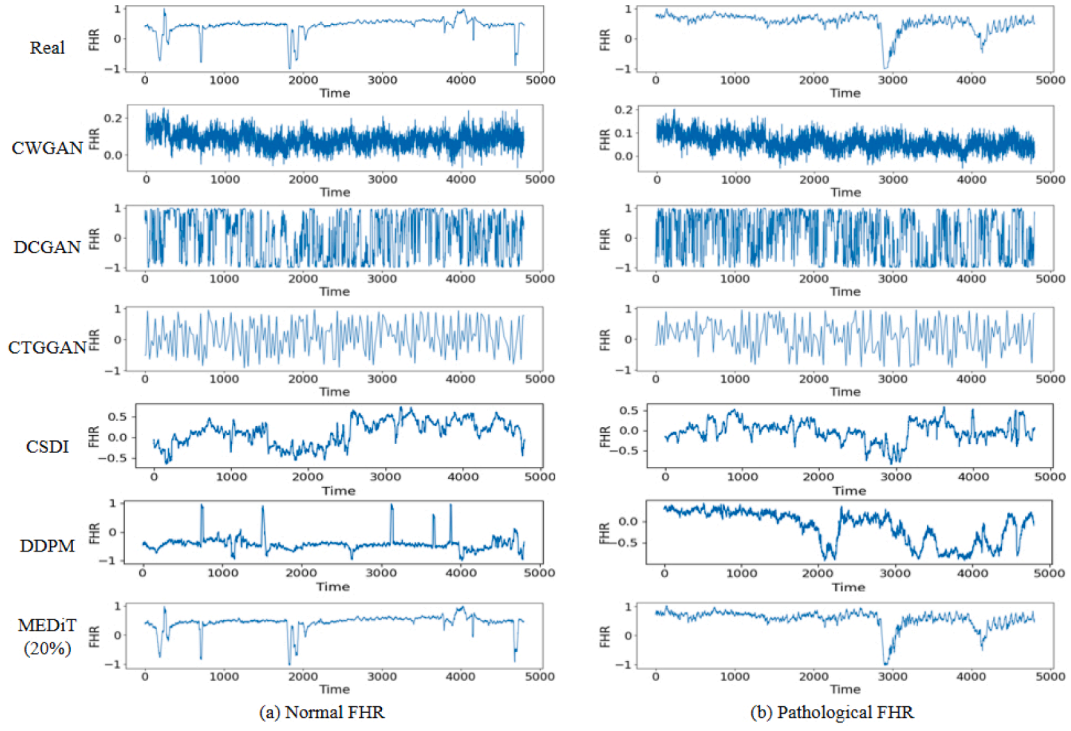


Fig. 8. Comparison of signals generated by various generative models.

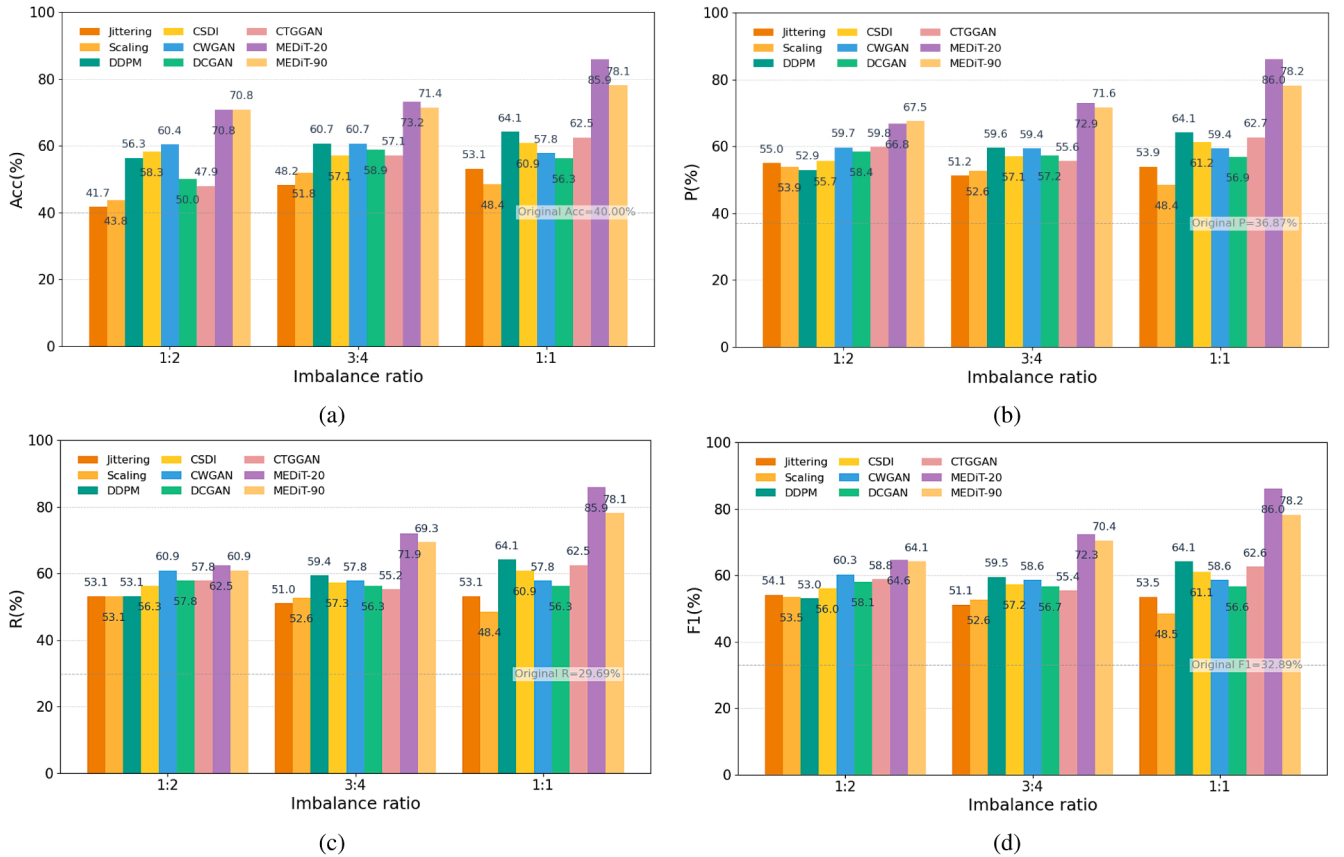


Fig. 9. Comparison of classification performance across different DA methods: (a) Accuracy; (b) Precision; (c) Recall; and (d) F1-score.

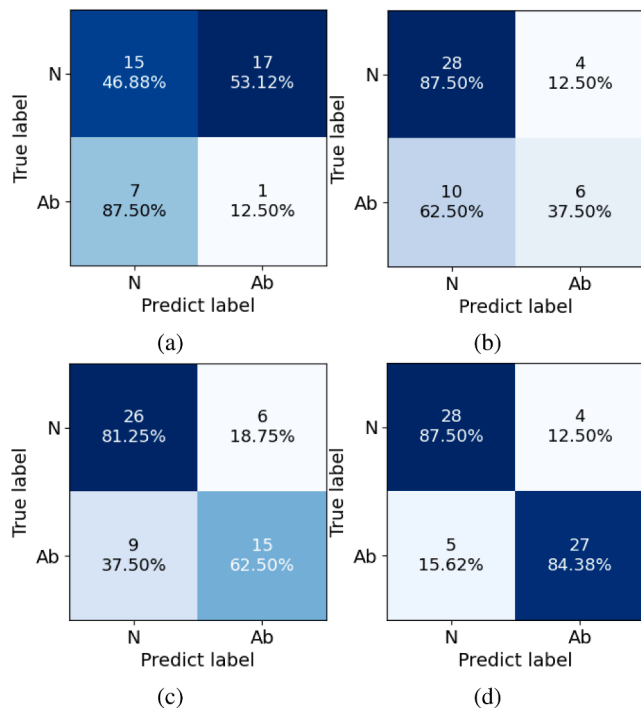


Fig. 10. Optimal classification confusion matrices across varying proportions of normal and abnormal signals: (a) Normal:Abnormal = 1:4 (Original); (b) Normal:Abnormal = 1:2; (c) Normal:Abnormal = 3:4; (d) Normal:Abnormal = 1:1.

At the original 1:4 imbalance ratio (without synthetic data), the baseline classifier achieved only 40% accuracy. When 40 synthetic pathological data were added (ratio 1:2), traditional augmentation methods and deep learning-based methods achieved comparable performance improvements. As the number of synthetic data increased to 80–120 (ratios of 3:4 to 1:1), diffusion-based methods began to outperform GAN-based approaches. MEDiT-20 achieved an accuracy of 85.9%, significantly surpassing the suboptimal CTGAN model (by 23.4%). This improvement can be attributed to the richer diversity of diffusion-generated samples, which better capture the underlying data distribution. Both MEDiT-20 and MEDiT-90 exhibited superior performance, achieving F1 of 86.6% and 78.2%, respectively. The comparable performance between both masking configurations further validates MEDiT's robustness across diverse missing-data conditions.

Fig. 10 illustrates the corresponding confusion matrices of MEDiT-20 under varying ratios between normal and abnormal samples. The results indicate that progressively incorporating synthetic abnormal data into the training set consistently enhances classifier performance across all metrics. Notably, MEDiT substantially reduces the false negative rate, which is a critical metric for clinical pathological diagnosis—from 53.12% (baseline) to 12.5% (1:1 ratio). This reduction is particularly important as missed diagnosis in clinical practice could lead to severe consequences, including delayed interventions and adverse neonatal outcomes.

5. Conclusion and future work

Obtaining sufficient high-quality FHR signals remains challenging because of ethical restrictions, privacy concerns, and acquisition artifacts. This paper proposed MEDiT, a mask-enhanced diffusion transformer for generating realistic FHR signals under diverse missing-data conditions. It combines a mask-enhanced reinforcement mechanism and a dual-branch learning strategy within a diffusion-based framework, enabling robust modeling across various missingness levels. Specifically, MERM guides the model to focus on clinically meaningful masked re-

gions, promoting the learning of intrinsic physiological patterns. Meanwhile, DBLS adaptively optimizes the model under different masking ratios, improving temporal representation learning and training robustness. Extensive experiments on the publicly CTU-UHB database show that MEDiT outperforms existing generative methods in terms of both signal fidelity and physiological plausibility. Furthermore, when applied to downstream classification tasks, MEDiT improves diagnostic accuracy by 14.0% over the best baseline method, DDPM, and significantly reduces false negatives, a clinically critical improvement that helps minimize missed diagnoses.

Overall, MEDiT provides a novel and effective solution for realistic FHR data augmentation in data-scarce scenarios, thereby supporting reliable computer-aided decision-support systems. Nevertheless, data augmentation should be viewed as a supportive method rather than a replacement for real clinical data, and further evaluation of privacy, ethical use, and cross-dataset generalizability is still needed before clinical deployment. Future work will extend this framework to a broader range of medical signal modalities and explore adaptive diffusion strategies guided by medical priors to further improve data realism, model interpretability, and clinical applicability.

CRedit authorship contribution statement

Yefei Zhang: Writing – review & editing, Writing – original draft, Software, Methodology, Funding acquisition, Conceptualization; **Junhao Hu:** Writing – review & editing, Writing – original draft, Validation; **Pengfei Jiao:** Methodology, Funding acquisition, Conceptualization; **Yanjun Deng:** Writing – review & editing, Software, Data curation; **Yehui Chen:** Data curation; **Zhidong Zhao:** Writing – review & editing, Methodology, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgment

This work was supported by the Natural Science Foundation of China under Grant No.62301205, the Zhejiang Provincial Natural Science Foundation of China under Grant No.LQ24F010011, the [Fundamental Research Funds for the Provincial Universities of Zhejiang](#) under Grant No. GK2599 09299001-021, and the Zhejiang Provincial Key Laboratory for Sensitive Data Security Protection and Confidentiality Management No. 2024E10048.

References

- Abdullah, A.A., Mohammed, N.S., Khanzadi, M., Asaad, S.M., Abdul, Z.K., & Maghddid, H.S. (2025). In-depth analysis on machine learning approaches: Techniques, applications, and trends. *Aro-The Scientific Journal of Koya University*, 13(1), 190–202.
- Alcaraz, J.M.L., & Strodthoff, N. (2022). Diffusion-based time series imputation and forecasting with structured state space models. *arXiv:2208.09399*
- Amer-Wählin, I., Hellsten, C., Norén, H., Hagberg, H., Herbst, A., Kjellmer, I., Lilja, H., Lindoff, C., Månsson, M., Mårtensson, L. et al. (2001). Cardiotocography only versus cardiotocography plus ST analysis of fetal electrocardiogram for intrapartum fetal monitoring: A swedish randomised controlled trial. *The Lancet*, 358(9281), 534–538.
- Awila, K.M., Veisi, H., & Abdullah, A.A. (2025). Sentiment analysis in low-resource contexts: BERT's impact on central kurkish. *Language Resources and Evaluation*, 59(3), 2213–2243.
- Bedin, L., Cardoso, G., Duchateau, J., Dubois, R., & Moulines, E. (2024). Leveraging an ECG beat diffusion model for morphological reconstruction from indirect signals. *Advances in NeurIPS*, 37, 84409–84446.
- Blattmann, A., Dockhorn, T., Kulal, S., Mendelevitch, D., Kilian, M., Lorenz, D., Levi, Y., English, Z., Voleti, V., Letts, A. et al. (2023). Stable video diffusion: Scaling latent video diffusion models to large datasets. *arXiv:2311.15127*
- Chandrasekaran, E., Pereira, S., Ghi, T., Perez-Bonfils, A.G., Fieni, S., Jia, Y.-J., Griffiths, K., Sukumaran, S., Ingram, C., Reeves, K. et al. (2024). International expert consensus statement on physiological interpretation of cardiotocograph (CTG): First revision (2024). *European Journal of Obstetrics & Gynecology and Reproductive Biology*, 302, 346–355.

- Cnattingius, S., Källén, K., Sandström, A., Rydberg, H., Månsson, H., Stephansson, O., Frisell, T., & Ludvigsson, J.F. (2023). The swedish medical birth register during five decades: Documentation of the content and quality of the register. *European Journal of Epidemiology*, 38(1), 109–120.
- Feng, J., Liang, J., Qiang, Z., Li, X., Chen, Q., Liu, G., Hong, J., Hao, Z., & Wei, H. (2021). Effective techniques for intelligent cardiocography interpretation using XGB-RF feature selection and stacking fusion. In *BIBM IEEE*. (pp. 2667–2673).
- Giuffrè, M., & Shung, D.L. (2023). Harnessing the power of synthetic data in healthcare: Innovation, application, and privacy. *NPJ Digital Medicine*, 6(1), 186.
- Goldberger, A.L., Amaral, L. A.N., Glass, L., Hausdorff, J.M., Ivanov, P.C., Mark, R.G., Mietus, J.E., Moody, G.B., Peng, C.-K., & Stanley, H.E. (2000). Physiobank, physiotoolkit, and physionet: Components of a new research resource for complex physiologic signals. *Circulation*, 101(23), e215–e220.
- Gregor, K., Papamakarios, G., Besse, F., Buesing, L., & Weber, T. (2018). Temporal difference variational auto-encoder. *arXiv:1806.03107*
- Gu, J., Shen, Y., Zhai, S., Zhang, Y., Jaitly, N., & Susskind, J. (2024). Kaleido diffusion: Improving conditional diffusion models with autoregressive latent modeling. *Advances in NeurIPS*, 37, 5498–5527.
- Ho, J., Jain, A., & Abbeel, P. (2020). Denoising diffusion probabilistic models. *Advances in NeurIPS*, 33, 6840–6851.
- Huang, H., Chen, M., & Qiao, X. (2024). Generative learning for financial time series with irregular and scale-invariant patterns. In *ICLR*.
- Kobyzev, I., Prince, S.J.D., & Brubaker, M.A. (2020). Normalizing flows: An introduction and review of current methods. *IEEE transactions on pattern analysis and machine intelligence*, 43(11), 3964–3979.
- Kong, Z., Ping, W., Huang, J., Zhao, K., & Catanzaro, B. (2021). DiffWave: A versatile diffusion model for audio synthesis. In *ICLR*.
- Kwak, J., & Jung, J. (2024). Classification of imbalanced ECGs through segmentation models and augmented by conditional diffusion model. *PeerJ Computer Science*, 10, e2299.
- Li, X., Lu, Y., Fu, X., & Qi, Y. (2021). Building the internet of things platform for smart maternal healthcare services with wearable devices and cloud computing. *Future Generation Computer Systems*, 118, 282–296.
- Li, X., Sakevych, M., Atkinson, G., & Metsis, V. (2024). BioDiffusion: A versatile diffusion model for biomedical signal synthesis. *Bioengineering*, 11(4), 299.
- Li, Y., Lu, X., Wang, Y., & Dou, D. (2022). Generative time series forecasting with diffusion, denoise, and disentanglement. *Advances in NeurIPS*, 35, 23009–23022.
- Liu, M., Lu, Y., Long, S., Bai, J., & Lian, W. (2021). An attention-based CNN-biLSTM hybrid neural network enhanced with features of discrete wavelet transformation for fetal acidosis classification. *Expert Systems with Applications*, 186, 115714.
- Lugmayr, A., Danelljan, M., Romero, A., Yu, F., Timofte, R., & Van Gool, L. (2022). Repaint: Inpainting using denoising diffusion probabilistic models. In *CVPR* (pp. 11461–11471).
- Luo, Y., Zhu, L., Wan, Z., & Lu, B. (2020). Data augmentation for enhancing EEG-based emotion recognition with deep generative models. *Journal of Neural Engineering*, 17(5), 056021.
- Marques, J.A.L., Han, T., Wu, W., do Vale Madeiro, J.P., Neto, A.V.L., Gravina, R., Fortino, G., & de Albuquerque, V.H.C. (2020). IoT-based smart health system for ambulatory maternal and fetal monitoring. *IEEE Internet of Things Journal*, 8(23), 16814–16824.
- Melamed, N., Baschat, A., Yinon, Y., Athanasiadis, A., Mecacci, F., Figueras, F., Berghella, V., Nazareth, A., Tahlak, M., McIntyre, H.D. et al. (2021). Figo (international federation of gynecology and obstetrics) initiative on fetal growth: Best practice advice for screening, diagnosis, and management of fetal growth restriction. *International Journal of Gynaecology and Obstetrics*, 152(Suppl 1), 3.
- Mirza, M., & Osindero, S. (2014). Conditional generative adversarial nets. *arXiv:1411.1784*
- PhysioNet (2014). CTU-CHB intrapartum cardiocography database. Accessed: May 2, 2025. <https://www.physionet.org/physiobank/database/ctu-uhb-ctgdb/>.
- Rasul, K., Seward, C., Schuster, L., & Vollgraf, R. (2021). Autoregressive denoising diffusion models for multivariate probabilistic time series forecasting. In *ICML* (pp. 8857–8868).
- Rombach, R., Blattmann, A., Lorenz, D., Esser, P., & Ommer, B. (2022). High-resolution image synthesis with latent diffusion models. In *CVPR* (pp. 10684–10695).
- Saharia, C., Chan, W., Chang, H., Lee, C., Ho, J., Salimans, T., Fleet, D., & Norouzi, M. (2022). Palette: Image-to-image diffusion models. In *SIGGRAPH* (pp. 1–10).
- Sang, Y., Beetz, M., & Grau, V. (2022). Generation of 12-lead electrocardiogram with subject-specific, image-derived characteristics using a conditional variational autoencoder. In *ISBI IEEE*. (pp. 1–5).
- Schirrmeyer, R.T., Springenberg, J.T., Fiederer, L.D.J., Glasstetter, M., Eggenberger, K., Tangermann, M., Hutter, F., Burgard, W., & Ball, T. (2017). Deep learning with convolutional neural networks for EEG decoding and visualization. *Human Brain Mapping*, 38(11), 5391–5420.
- Shen, L., Chen, W., & Kwok, J. (2024). Multi-resolution diffusion models for time series forecasting. In *ICLR*.
- Shen, L., & Kwok, J. (2023). Non-autoregressive conditional diffusion models for time series prediction. In *ICML* (pp. 31016–31029).
- Sohn, K., Lee, H., & Yan, X. (2015). Learning structured output representation using deep conditional generative models. *Advances in NeurIPS*, 28, 3483–3491.
- Suh, S., Lee, H., Lukowicz, P., & Lee, Y. O. (2021). CEGAN: Classification enhancement generative adversarial networks for unraveling data imbalance problems. *Neural Networks*, 133, 69–86.
- Taha, Z.Y., Abdullah, A.A., & Rashid, T.A. (2025). Optimizing feature selection with genetic algorithms: a review of methods and applications. *Knowledge and Information Systems*, 67(11), 9739–9778.
- Tashiro, Y., Song, J., Song, Y., & Ermon, S. (2021). CSDI: Conditional score-based diffusion models for probabilistic time series imputation. *Advances in NeurIPS*, 34, 24804–24816.
- Torma, S., & Szegletes, L. (2025). Generative modeling and augmentation of EEG signals using improved diffusion probabilistic models. *Journal of Neural Engineering*, 22(1), 016001.
- Wang, Y., Zhao, S., Jiang, H., Li, S., Luo, B., Li, T., & Pan, G. (2024). DiffMDD: A diffusion-based deep learning framework for MDD diagnosis using EEG. *IEEE Transactions on Neural Systems and Rehabilitation Engineering*, 32, 728–738.
- Wang, Z., Zheng, H., He, P., Chen, W., & Zhou, M. (2023). Diffusion-GAN: Training GANs with diffusion. In *ICLR*.
- Yousefimehr, B., Ghatte, M., Seifi, M.A., Fazli, J., Tavakoli, S., Rafei, Z., Ghaffari, S., Nikahd, A., Gandomani, M. R., Orouji, A. et al. (2025). Data balancing strategies: A survey of resampling and augmentation methods. *arXiv:2505.13518*
- Yu, S., Nie, W., Huang, D.-A., Li, B., Shin, J., & Anandkumar, A. (2024a). Efficient video diffusion models via content-frame motion-latent decomposition. In *ICLR*.
- Yu, S., Sohn, K., Kim, S., & Shin, J. (2023). Video probabilistic diffusion models in projected latent space. In *CVPR* (pp. 18456–18466).
- Yu, X., Zhang, X., Cao, Y., & Xia, M. (2019). VAEGAN: A collaborative filtering framework based on adversarial variational autoencoders. In *IJCAI* (vol. 19). (pp. pp–4206).
- Yu, Z., Hu, Y., Lu, Y., Li, L., Ge, H., & Fu, X. (2024b). CTGGAN: Reliable fetal heart rate signal generation using gans. In *ICNN IEEE*. (pp. 1–8).
- Zhang, A., Su, L., Zhang, Y., Fu, Y., Wu, L., & Liang, S. (2022a). EEG data augmentation for emotion recognition with a multiple generator conditional wasserstein gan. *Complex & Intelligent Systems*, 8(4), 3059–3071.
- Zhang, J., Huang, Q., Liu, J., Guo, X., & Huang, D. (2025). Diffusion-4k: Ultra-high-resolution image synthesis with latent diffusion models. In *CVPR* (pp. 23464–23473).
- Zhang, Y., Zhao, Z., Deng, Y., & Zhang, X. (2022b). FHRGAN: Generative adversarial networks for synthetic fetal heart rate signal generation in low-resource settings. *Information Sciences*, 594, 136–150.
- Zhang, Y., Zhao, Z., Deng, Y., Zhang, X., & Zhang, Y. (2021). Reconstruction of missing samples in antepartum and intrapartum FHR measurements via mini-batch-based minimized sparse dictionary learning. *IEEE Journal of Biomedical and Health Informatics*, 26(1), 276–288.